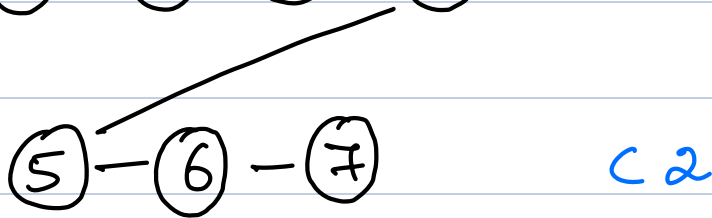
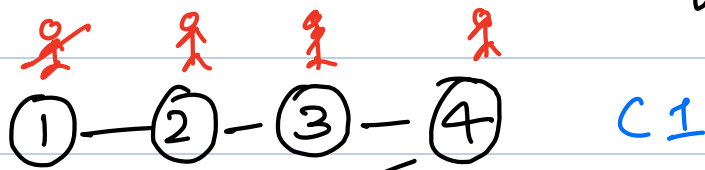


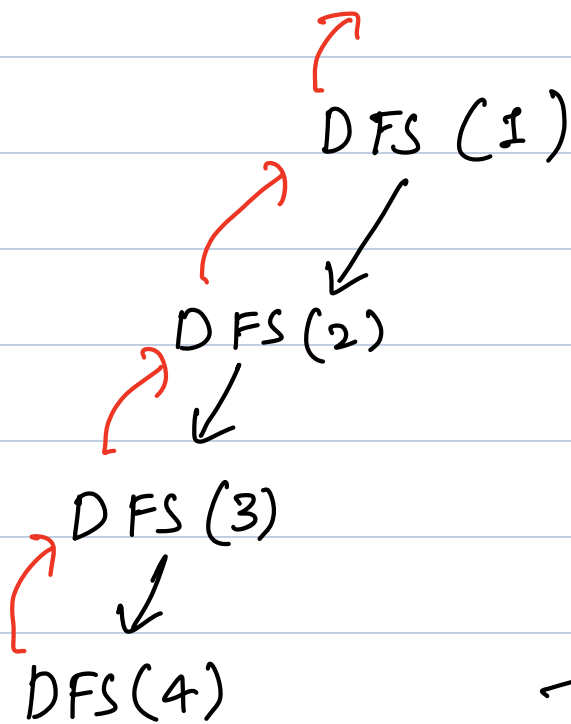
* Disjoint Set [Graph Algorithm]



7 & 1 $\Rightarrow$ No

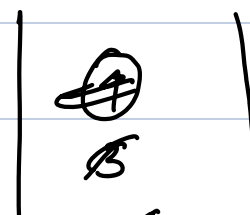
BFS / DFS

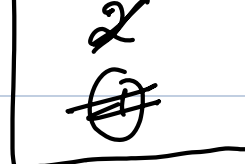
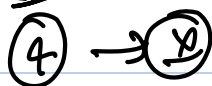
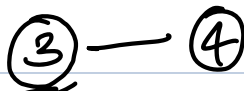
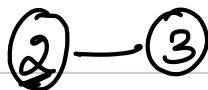
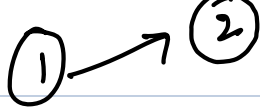
1	2	3	4	5	6	7
1 0	1 0	0 1	0 1	0	0	0



Time Complexity = $O(V+E)$

BFS





$$O(V+E)$$

* Disjoint Set $\Rightarrow$ $O(1)$

1, 2

3, 4

2, 3

5, 6 ✓

6, 7

4, 6 ✓



① find the parent (ultimate parent)

② Union $\left[\begin{array}{l} \text{Size} \\ \text{Rank} \end{array} \right.$

	0	1	2	3	4	5	6	7
Rank	0	1	0	1	0	1	0	0
		1		1				
		1	2	3		5	6	
Parent	0	1	2	3	4	5	6	7
	0	1	2	3	<u>4</u>	5	6	7

$$Up[u]^1 = 1$$

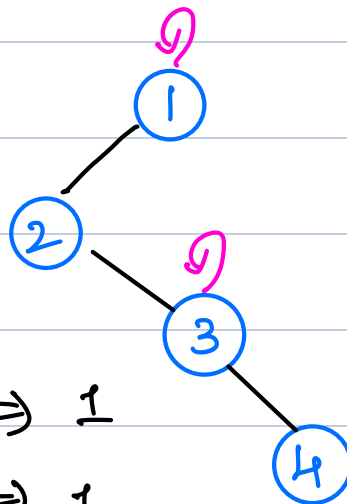
$$Up[v]^2 = 2$$

$$U = 1$$

$$V = 2$$

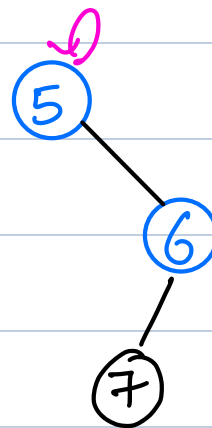
$$Rank[1] = 0$$

$$Rank[2] = 0$$



$$2 \Rightarrow 1 \Rightarrow 1$$

$$3 \Rightarrow 3 \Rightarrow 1$$



$$6 \Rightarrow 5 \Rightarrow 1$$

$$7 \Rightarrow 7 \Rightarrow 0$$

$$\log(N)$$

$$4 \Rightarrow 1 \Rightarrow 2$$

$$6 \Rightarrow 5 = 1$$

741

$up[7] == up[1] = \text{Yes}$

No
E

```
class DisjointSet {
    vector<int> Rank, Parent;
```

DisjointSet(int n) {

Rank.resize(n, 0);

Parent - re Size (n);

```
for (i=0; i<=n; i++)
```

Parent [i] = i;

3: 3

```
int findparent (int node) {
```

```
if (Parent[node] == node) {
```

return 1 node

Prent [node] = find parent (Parent [node])

2

Void Union by Rank (int u, int v) {

int pu = findparent(u);

int pv = findparent(v);

if (Rank[pu] > Rank[pv]) {

Parent[v] = u;

}

else if (Rank[pu] < Rank[pv]) {

Parent[u] = v;

}

else {

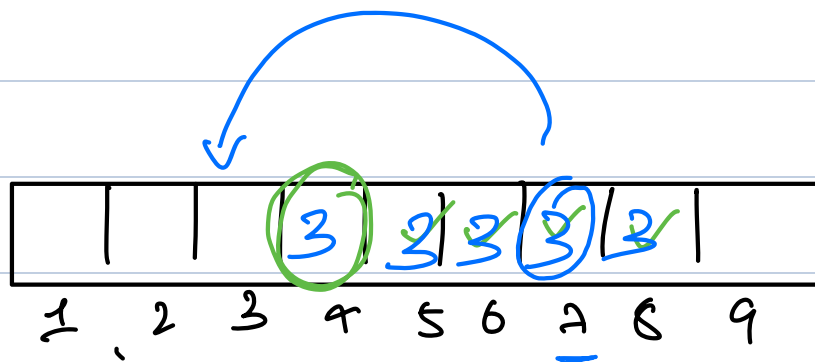
Parent[v] = u;

Rank[pu] ++;

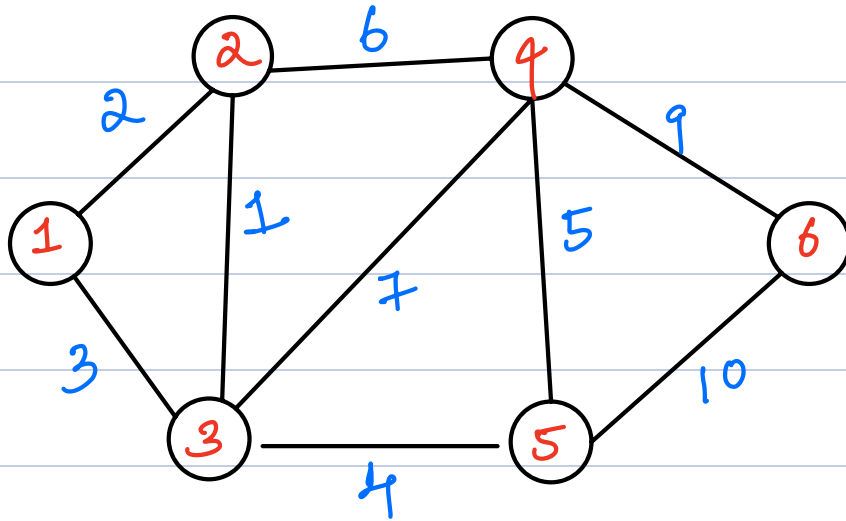
}

}

⑧

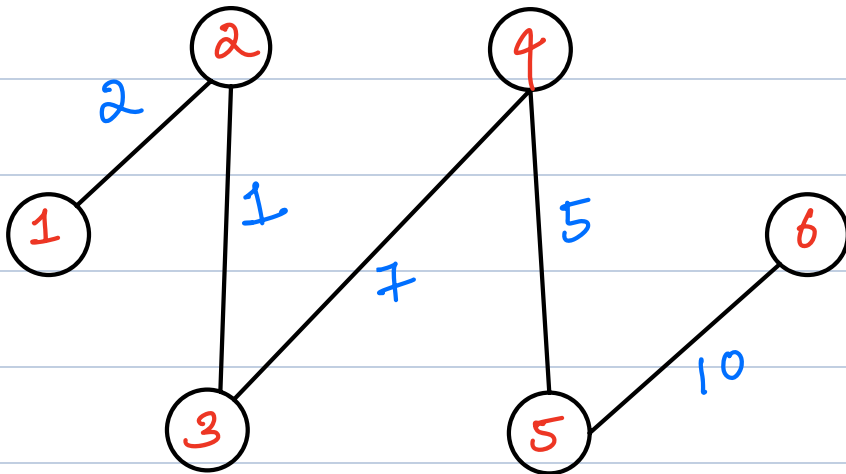


* Greedy [MST]

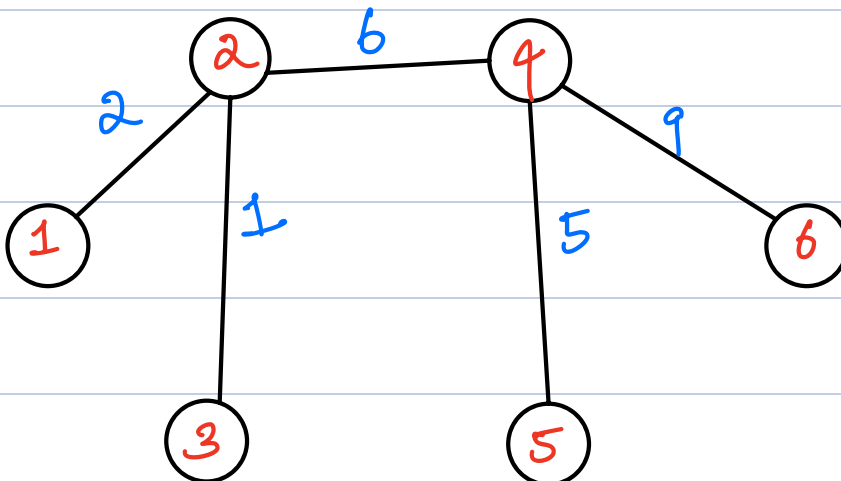


$n \rightarrow$ nodes

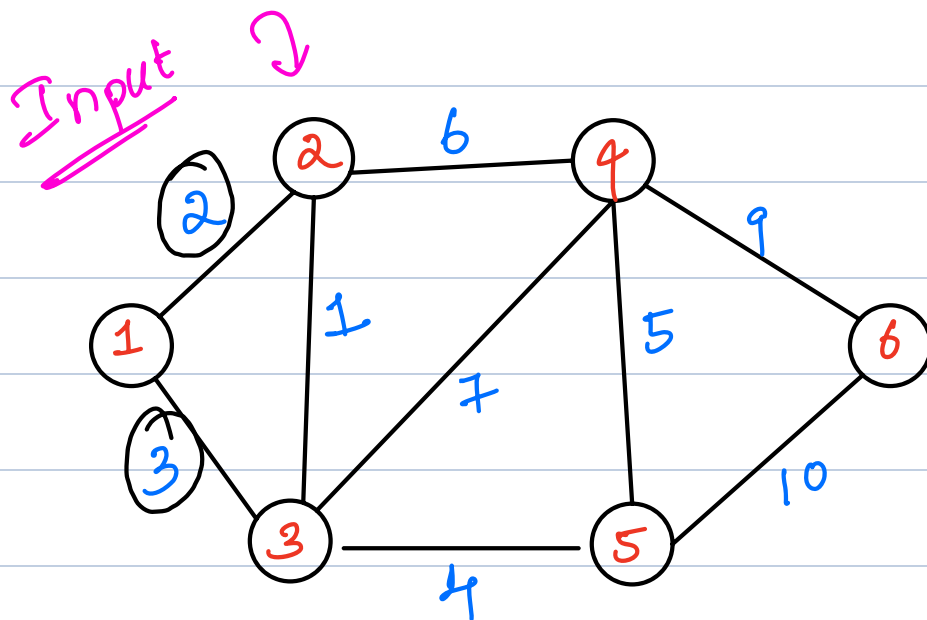
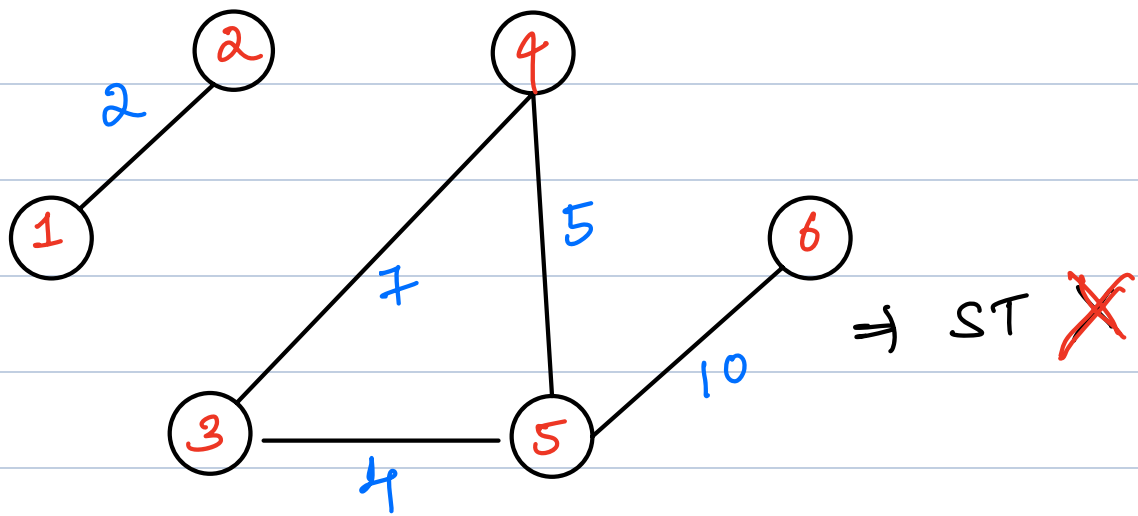
$(n-1) \Rightarrow$ Edges.



$\Rightarrow$ ST ✓



$\Rightarrow$ ST ✓

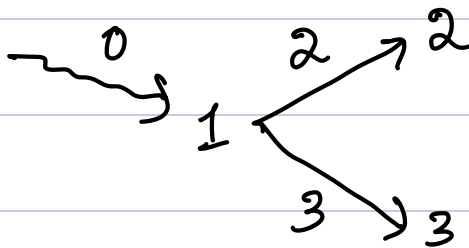
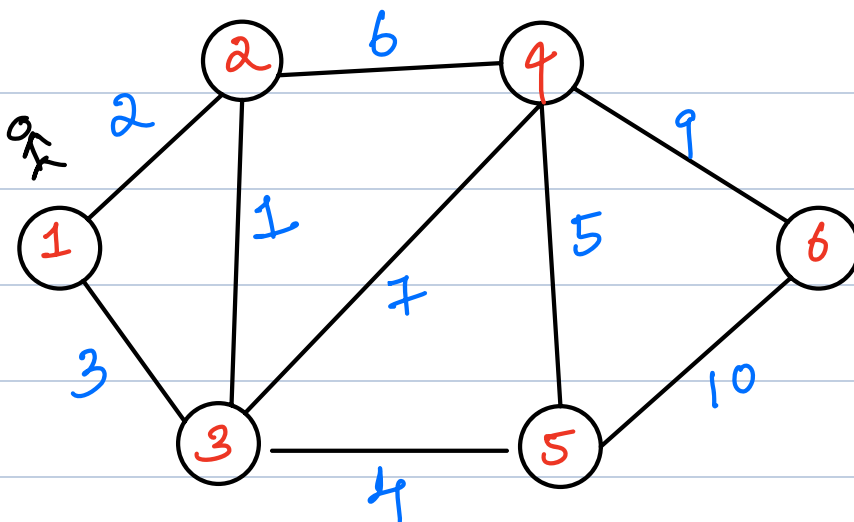
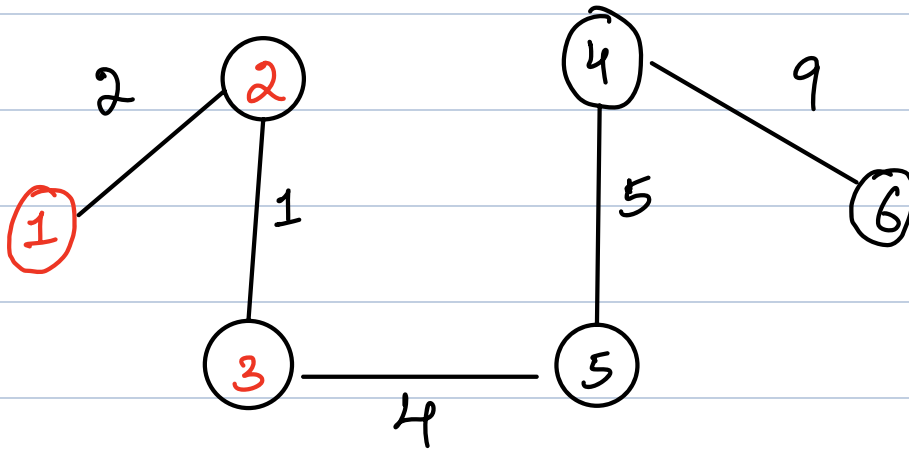


u	v	wt
✓1	2	2
✓1	3	3
✓2	3	1
2	4	6
✓3	5	4
✓4	5	5
4	6	9
5	6	10

Sort
⇒
wt

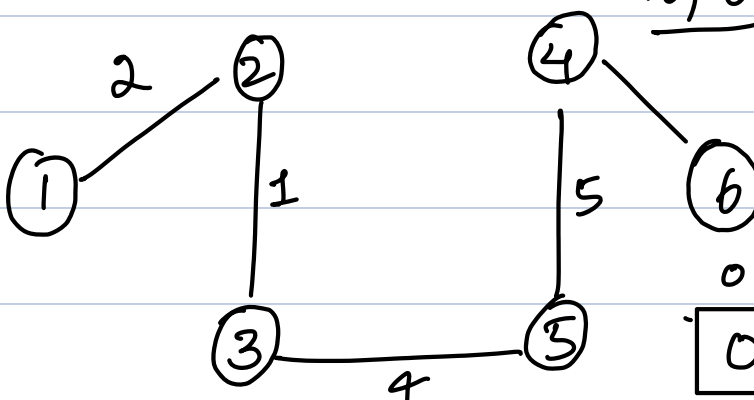
u	v	wt
2	3	1
1	2	2
✗1	3	3
✓3	5	4
✗4	5	5
✗2	4	6
4	6	9
5	6	10

0	1	2	3	4	5	6
0	0	0	0	0	0	0
	<u>1</u>	<u>1</u>	1	<u>1</u>	<u>1</u>	



9, 6
 5, 4 ✓
 10, 6 ✓

wt, Node
0, 1
2, 2
<u>3, 3</u>
1, 3
6, 4 ✓
<u>4, 5</u>
7, 4 ✓



0	1	2	3	4	5	6
0	0	0	0	0	0	0

1 1 1 1 1

0	
1	$\{\{2, 2\}, \{3, 3\}\}$
2	$\{\{1, 2\}\}$
3	$\{\{1, 3\}\}$
4	
5	
6	

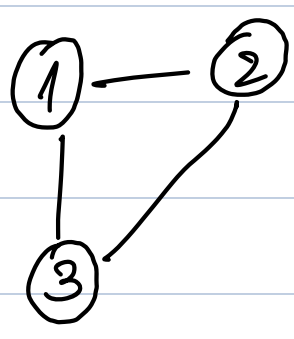
$\Rightarrow$ weighted Graph

$\text{Vector} \langle \text{pair} \langle \text{int}, \text{int} \rangle \rangle \text{adj} [n+1];$

u	v
1	2
1	3
2	3

$\Rightarrow$

1	$\{2, 3\}$
2	$\{1, 3\}$
3	$\{1, 2\}$



$\text{Vector} \langle \text{int} \rangle \text{adj} [n+1]$